Advanced Quantitative Research Methods

Causal Inference from Observational Data

Advanced module A10 — optional self-study

Prof. Dr. Christoph Weisser

Slides, lab notebooks and data:
github.com/ChrisW09/Advanced-Quantitative-Research-Methods

Where this module fits

Course chapter / module	What this module builds on it
Ch. 3 — linear regression	Adjustment *is* multiple regression; omitted-variable bias
Ch. 4 — logistic regression	The propensity score is a fitted logistic model
Ch. 5 — resampling	Seeded simulations that know the true effect
Module A1 — RCTs	Potential outcomes, selection bias — assumed throughout

The question A1 left open

Module A1 ended with the one clean design: randomise, then compare. This module asks what you can still learn when **nobody randomised** — the data are observational, treatment was *chosen*, and every naive comparison is a mixture of effect and selection.

One of twelve optional advanced modules; self-study. Every number on these slides is reproduced by `make_figures.py` in the module folder.

Contents

1. The Problem
2. Omitted-Variable Bias
3. Adjustment & Matching
4. Difference-in-Differences
5. Instrumental Variables
6. Python Lab
7. Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
$D \in \{0, 1\}$	treatment received (not assigned — nobody assigned it)
$Y(1), Y(0)$	potential outcomes with and without treatment (module A1)
U	a confounder: causes both D and Y
X	observed covariates (possibly a noisy proxy for U)
$e(x) = \Pr(D = 1 \mid X = x)$	the propensity score
Z	an instrument: shifts D, touches Y only through D
τ	the causal effect we are after (ATT/ATE)
δ	auxiliary-regression coefficient of D on the confounder

Sources and references

Primary textbook

James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning, with Applications in Python*, Springer. — The regression and classification machinery this module reuses.

Canonical further reading (optional)

Angrist & Pischke, *Mostly Harmless Econometrics* (2009); Hernán & Robins, *Causal Inference: What If* (2020) — both develop everything here at full depth.

Learning objectives

By the end of this module you will be able to:

- **Explain** why a naive group comparison in observational data mixes the causal effect with selection bias.
- **Derive** the omitted-variable-bias formula and predict the sign and size of the bias.
- **Adjust** for observed confounders by regression, stratification and propensity-score matching — and say what adjustment can never fix.
- **Apply** difference-in-differences to a two-group, two-period table and state the parallel-trends assumption.
- **Use** an instrumental variable via the Wald estimator, and check the three conditions an instrument must satisfy.

Roadmap of this module

From "the data say" to "the design says"

1. The problem: selection, confounding, and one running simulation.
2. Omitted-variable bias: the formula behind every headline gone wrong.
3. Adjustment: regression, stratification, propensity-score matching.
4. Difference-in-differences: using time as the control.
5. Instrumental variables: borrowing someone else's randomness.
6. Python lab, summary, appendix.

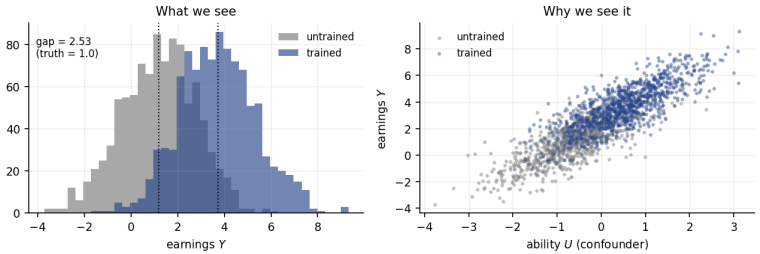
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The running example: a job-training programme

A firm offers a voluntary training programme. Ability U drives both **who signs up** (D) and **later earnings** (Y):

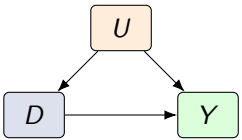
$$Y = 2 + \underbrace{1.0}_{\text{true effect}} \cdot D + 1.5\,U + \varepsilon, \quad D = 1\{0.9\,U + \eta > 0\}, \quad n = 2000.$$



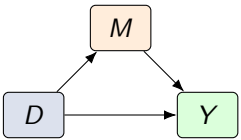
The gap is not the effect

The naive difference in means is 2.53; the true effect is 1.0. The right panel shows why: the trained were more able *before* training.

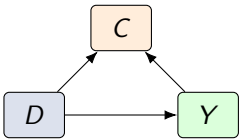
Confounder, mediator, collider — three roles, one picture



confounder: adjust



mediator: do *not* adjust



collider: do *not* adjust

How to read this

Arrows are causes. A **confounder** sits *upstream* of both D and Y and must be adjusted away. A **mediator** carries the effect — adjusting for it deletes the very thing you measure. A **collider** is caused by both — adjusting for it *creates* spurious association. So "control for everything" is not a strategy: what to adjust for is a *causal* decision, made from the diagram, not from p-values.

Exercise A10.1 — Adjust, or leave alone? [Concept]

Exercise

You study the effect of **smoking** (D) on **heart disease** (Y). For each variable, say whether adjusting for it helps, is harmless, or actively biases the estimate — and name its role:

1. **Age** — older people smoke more (in this cohort) and have more heart disease.
2. **Tar deposits in the lungs** — caused by smoking; contribute to disease.
3. **Hospitalisation** — both heavy smokers and heart patients are over-represented in a sample recruited *at a hospital*.

Solution — Exercise A10.1

Solution

Method. Place each variable in the diagram relative to D and Y; the position decides.

1. **Age: confounder** (Age $\to D$, Age $\to Y$). Adjust — without it the smoking effect absorbs the age effect.
2. **Tar: mediator** ($D \to$ Tar $\to Y$). Do *not* adjust: comparing smokers and non-smokers *at the same tar level* removes the pathway through which smoking does its damage, biasing the total effect toward zero.
3. **Hospitalisation: collider** ($D \to H \leftarrow Y$). Recruiting only hospital patients *conditions* on H and manufactures a (typically negative) spurious association — smokers in hospital are there partly *because* they smoke, patients without heart disease partly because of other illness.

Common mistake

Adding every available covariate to the regression "to be safe". Item 2 and item 3 are both cases where more adjustment gives a *worse* answer.

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The omitted-variable-bias formula

Suppose the truth is the **long** regression, but you fit the **short** one:

$$\text{long: } Y = \beta_0 + \tau D + \gamma U + \varepsilon \qquad \text{short: } Y = b_0 + b_D\, D + e$$

Rule

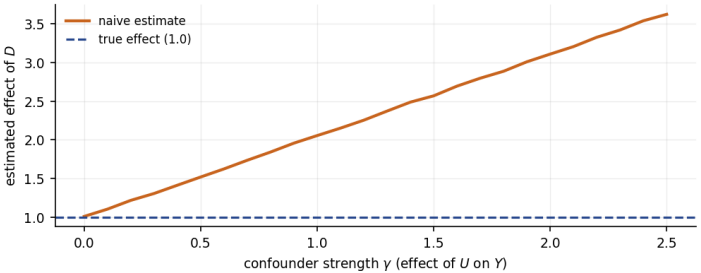
$$\boxed{b_D \;=\; \tau + \gamma\,\delta} \qquad \text{where } U = \delta_0 + \delta D + v \text{ (the auxiliary regression).}$$

How to read this

The bias is a *product*: how strongly the omitted variable drives the outcome (γ) $\times$ how different the groups are on it (δ). Either factor zero $\Rightarrow$ no bias. Both positive $\Rightarrow$ you *overstate* the effect.

Chapter 3 showed coefficients move when predictors enter; this formula says by exactly how much — for variables you cannot even observe.

The bias is linear in the confounder's strength



Read off the simulation

Dialling the confounder's effect γ from 0 to 2.5, the naive estimate climbs linearly with slope ≈ 1.05 — which *is* δ, the ability gap between the self-selected groups. At $\gamma = 1.5$: $1.0 + 1.5 \times 1.05 \approx 2.5$, the naive gap we saw.

Exercise A10.2 — Sign and size the bias [Math]

Exercise

A study of **private tutoring** (D) on **exam scores** (Y, points) omits **parental income** U (in \$10k). Assume the long model $Y = \beta_0 + \tau D + \gamma U + \varepsilon$ with $\tau = 5$, $\gamma = 1.5$, and the auxiliary regression $U = \delta_0 + \delta D + v$ with $\delta = 0.8$.

1. What does the short regression of Y on D estimate?
2. A second omitted variable, **prior struggle**, has $\gamma = -2$ and $\delta = 1.2$ (struggling students get tutors). Its bias?
3. With *both* omitted, can you sign the total bias without a computer?

Solution — Exercise A10.2

Solution

Method. Apply $b_D = \tau + \sum_j \gamma_j \delta_j$, one omitted variable at a time; biases from independent confounders add.

1. $b_D = 5 + 1.5 \times 0.8 = 5 + 1.2 = 6.2$ — tutoring looks 24% better than it is, because richer families buy more of it.
2. Bias $= \gamma\delta = (-2)(1.2) = -2.4$ — struggling students select *into* tutoring, dragging the tutored group's scores down.
3. Total: $5 + 1.2 - 2.4 = 3.8$. The two selections push in *opposite* directions, and here the negative one wins — naive comparisons can just as easily *understate* an effect.

Take-away. You rarely know γ and δ; the formula's real use is a *sensitivity argument*: "how strong would a confounder have to be to explain my estimate away?"

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Strategy 1: regression adjustment — and its honest limit

Rule

If **all** confounders are observed as X (*unconfoundedness*), regressing Y on D *and* X recovers the effect: $\hat{\tau}$ = coefficient of D in $Y \sim D + X$.

The running simulation, adjusted

We observe only a noisy proxy $X = U + \text{noise}$. Adjusting for it moves the estimate from the naive 2.53 to 1.35 — much closer to the truth 1.0, but *not there*: the noise in the proxy leaves residual confounding behind.

The assumption does all the work

Adjustment removes exactly the bias carried by **what you measured**. The claim "no unmeasured confounders" is not testable from the data — it is an argument you must make about the world.

Strategy 2: stratify — compare like with like

The idea in one sentence

Cut the data into groups (strata) inside which X is essentially constant, compare treated and control *within* each stratum, then average the within-stratum gaps.

Why it works

Within a stratum, treated and control have the same X — so the $\gamma\delta$ term of the bias formula is switched off (for the measured X). The naive comparison is a *single* stratum: everyone at once.

Simpson's warning

The within-stratum gaps and the pooled gap can even have *opposite signs* — module A1's Simpson slide is exactly this phenomenon. Stratification is the cure, when the strata are confounders.

Exercise A10.3 — Stratification by hand [Math]

Exercise

A voluntary coaching programme; $X \in \{\text{low}, \text{high}\}$ prior skill. Mean outcomes (group sizes in parentheses):

	coached	**not coached**
low skill	5.0 $(n = 20)$	4.0 $(n = 80)$
high skill	9.0 $(n = 80)$	8.0 $(n = 20)$

1. Compute the naive difference in means (pool the strata).
2. Compute the stratified estimate (average the within-stratum gaps, weighting each stratum by its total size).
3. Explain the gap between the two numbers in one sentence.

Solution — Exercise A10.3

Solution

Method. Naive = overall treated mean − overall control mean; stratified = size-weighted average of within-stratum gaps.

Part 1 — naive. Treated: $(20 \times 5 + 80 \times 9)/100 = 8.2$. Control: $(80 \times 4 + 20 \times 8)/100 = 4.8$. Naive gap $= 8.2 - 4.8 = 3.4$.

Part 2 — stratified. Within low: $5 - 4 = 1$. Within high: $9 - 8 = 1$. Both strata hold 100 people, so the weighted average is 1.0.

Part 3. The coached are mostly high-skill (80 of 100) and the uncoached mostly low-skill, so the naive gap adds the *skill* gap (≈ 4 points, diluted to 2.4) on top of the *coaching* gap (1).

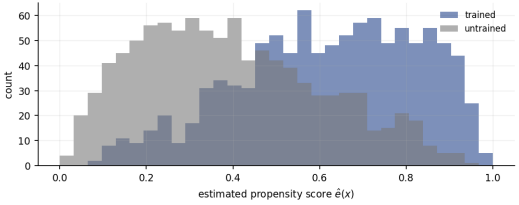
Common mistake

Weighting the within-stratum gaps by the number of *treated* only when you want the ATE — that weighting answers the ATT question instead (here both give 1.0 because the gaps are equal; in general they differ).

Strategy 3: the propensity score — one number replaces all of X

Rule (Rosenbaum & Rubin)

Let $e(x) = \Pr(D = 1 \mid X = x)$, fitted with Chapter 4's logistic regression. If unconfoundedness holds given X, it also holds given the *scalar* $e(X)$ — so it is enough to compare units with the same propensity score.

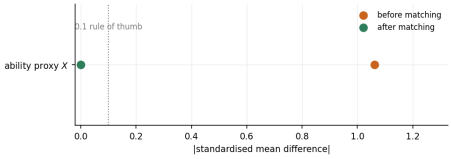


Check overlap before anything else

Matching needs treated and control units at *comparable* scores. Where one histogram is empty, no method can compare — those units have no counterpart.

Propensity-score matching on the running example

Recipe (ATT): for each treated unit, find the control with the nearest $\hat{e}(x)$ (with replacement); average the within-pair differences.



What matching bought us

Before matching the groups differ by 1.06 standard deviations on the ability proxy; after matching, 0.00. The estimate moves from the naive 2.53 to 1.42 — the same story as regression adjustment (1.35), because both lean on the same assumption.

Common mistake

Reporting the matched estimate without the balance picture — the SMD-before/after plot **is** the evidence that matching did anything.

Adjustment scoreboard — one simulation, four answers

Estimator	Estimate	Why it lands there
Truth (built into the simulation)	1.00	—
Naive difference in means	2.53	effect + selection ($\gamma\delta$)
Regression on D and proxy X	1.35	removes measured confounding only
Propensity matching on X	1.42	same assumption, same residual gap

Adjustment methods agree with each other, not with the truth

Regression and matching answer the *same* question two ways — so their agreement is a consistency check, **not** a validation. Both still carry the confounding that the noisy proxy could not see.

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Using time as the control group

A policy hits one group (treated) and not another (control); you observe both **before** and **after**.

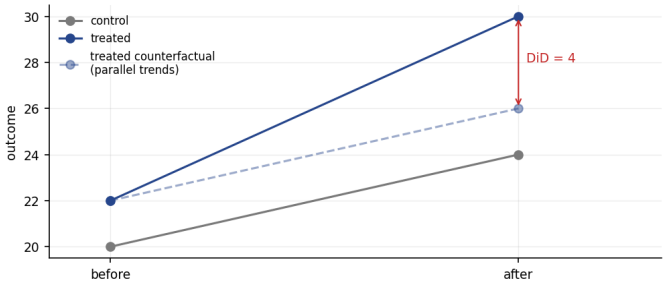
Rule

$$\hat{\tau}_{\text{DiD}} = \left(\bar{Y}_{\text{T,after}} - \bar{Y}_{\text{T,before}}\right) - \left(\bar{Y}_{\text{C,after}} - \bar{Y}_{\text{C,before}}\right)$$

How to read this

The control group's change estimates what would have happened to the treated *anyway* (time trend, seasonality, the economy). Subtracting it removes any confounder that is **constant over time within group** — even unmeasured ones. What it cannot remove: anything that changed for one group only.

DiD in one picture



Parallel trends is the whole bet

Control rose 20 $\rightarrow$ 24; treated rose 22 $\rightarrow$ 30; DiD $= 8 - 4 = 4$. The dashed line is the assumption: untreated, the treated would have risen by the control's $+4$. Plot pre-period trends — parallel *before* is the only evidence you will ever get.

Exercise A10.4 — DiD by hand [Math]

Exercise

A minimum-wage rise hits state T; neighbouring state C is untouched. Average monthly employment per fast-food store:

	before	**after**
state T (treated)	48	60
state C (control)	50	56

1. Compute the DiD estimate of the policy effect.
2. Compute the two *wrong* answers: after-minus-before for T only, and T-minus-C after only. What does each one confound the effect with?
3. Name one concrete event that would break parallel trends here.

Solution — Exercise A10.4

Solution

Part 1. $(60 - 48) - (56 - 50) = 12 - 6 = +6$ jobs per store.

Part 2. *Before-after in T only:* $+12$ — confounds the policy with the common time trend (the economy grew: C rose $+6$ with no policy at all). *T vs C after only:* $60 - 56 = +4$ — confounds the policy with the *permanent* level difference between the states (T started 2 below C, so this understates).

Part 3. Any asymmetric shock: a factory opening in state T between the waves, a tourism collapse hitting only C's border towns, T announcing the rise early so stores hired ahead. Parallel trends is a claim about *events*, not about data.

Common mistake

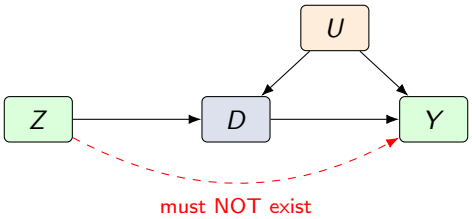
Treating DiD as assumption-free because it "differences everything out". It differences out *time-constant* confounding only — the two wrong answers in part 2 are each one of the two things it removes.

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Borrowing someone else's randomness

When a confounder is unmeasured and untimed, one door remains: a variable Z that *nudges* treatment but has no path of its own to the outcome.



The three conditions

1. **Relevance:** Z actually moves D (testable — the first stage).
2. **Exclusion:** Z affects Y *only* through D (not testable).
3. **Independence:** Z is as good as random — unrelated to U (not testable).

The Wald estimator — two small divisions

Rule (binary Z)

$$\hat{\tau}_{\text{IV}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

$$\frac{\text{effect of the nudge on the outcome}}{\text{effect of the nudge on take-up}}$$

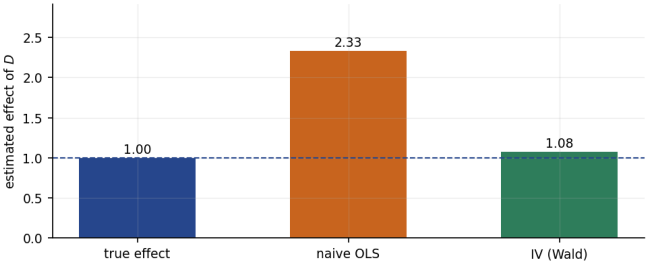
How to read this

The numerator is an *intention-to-treat* effect (module A1); the denominator is the **first stage**. Dividing scales the ITT up to the effect on those the nudge actually moved — the *compliers*. This is the same ITT logic A1 used for non-compliance, now run in reverse.

The running simulation, instrumented

A random encouragement Z raises take-up by 0.40 (first stage). Naive OLS: 2.33. Wald: 1.08 — on top of the truth 1.0, with the confounder U still completely unobserved.

Three estimators, one target



Why IV wins here — and what it costs

IV never observed the confounder, yet lands on the truth: the encouragement was *randomised*, so its variation in D carries no selection. The price: a **weak first stage** blows the variance up ($1/0.40$ scales every error), and the answer is the compliers' effect, not everyone's.

Exercise A10.5 — Is it an instrument? [Concept]

Exercise

You want the effect of **university attendance** (D) on **earnings** (Y). For each candidate Z, check relevance / exclusion / independence and give a verdict:

1. **Distance to the nearest university** at age 18.
2. **Parental education**.
3. **A lottery** that randomly awards tuition waivers among applicants.

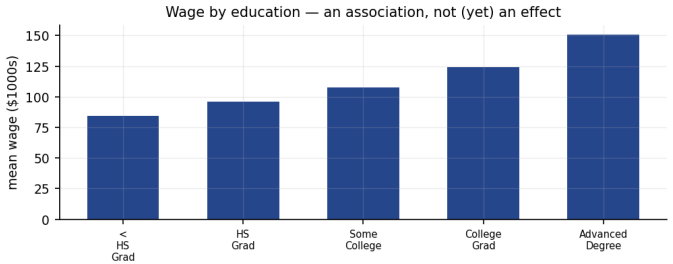
Solution — Exercise A10.5

Solution

Method. Walk the three conditions; one clear failure kills the candidate.

1. **Distance:** relevance plausible (nearer ⇒ more attendance — check the first stage). Exclusion is the worry: families near universities differ (urban, richer labour markets pay differently). Verdict: *usable only with controls for region — and arguable*. This is the classic Card (1995) design and the classic objection to it.
2. **Parental education:** relevant, yes — but it fails independence/exclusion completely: it drives ability, networks and expectations, all of which touch earnings directly. Verdict: *not an instrument; it is a confounder*.
3. **The lottery:** relevance testable (waiver ⇒ higher enrolment); independence holds by randomisation; exclusion plausible (a waiver moves earnings only via attendance — unless the cash itself matters). Verdict: *the best of the three by far*. Randomisation is what makes instruments credible — which is why the strongest IVs are embedded experiments.

Reality check: Wage data



What we can, and cannot, say

College graduates out-earn non-graduates by $36.3k; adjusting for age and year barely moves it ($35.1k). Nothing in this dataset measures ability or family background — so this module's honest verdict is: a large, robust *association*, whose causal share needs a design (a DiD, an instrument, a lottery) the data do not contain.

Extended Exercise A10.1 — Recover the truth in simulation [Python]

Extended exercise (15 min)

Simulate the running example yourself ($n = 2000$, seed 2024): $U \sim N(0,1)$, $X = U + N(0, 0.5)$, $D = 1\{0.9U + N(0,1) > 0\}$, $Y = 2 + D + 1.5U + N(0,1)$.

1. Compute the naive difference in means and the regression-adjusted estimate (on X). Compare to 1.0.
2. Re-run with $X = U$ exactly (no proxy noise). What happens to the adjusted estimate, and why?
3. Vary the proxy noise sd over $\{0, 0.25, 0.5, 1.0\}$ and plot the adjusted estimate against it. What does the curve tell you about "controlling for" badly measured confounders?

Solution — Extended Exercise A10.1 (1/2)

Solution — code

```
import numpy as np
rng = np.random.default_rng(2024)
n, out = 2000, {}
for s in [0, 0.25, 0.5, 1.0]: # proxy noise sd
    U = rng.normal(0, 1, n)
    X = U + rng.normal(0, s, n)
    D = (0.9*U + rng.normal(0, 1, n) > 0).astype(float)
    Y = 2 + D + 1.5*U + rng.normal(0, 1, n)
    naive = Y[D == 1].mean() - Y[D == 0].mean()
    Z = np.column_stack([np.ones(n), D, X])
    adj = np.linalg.lstsq(Z, Y, rcond=None)[0][1]
    out[s] = (naive, adj)
    print(f"noise sd {s}: naive {naive:.2f}, adjusted {adj:.2f}")
```

Solution — Extended Exercise A10.1 (2/2)

Solution — reading the output

Part 1. With noise sd 0.5 you reproduce the deck: naive ≈ 2.5, adjusted ≈ 1.3–1.4 (seed-dependent in the second decimal), truth 1.0.

Part 2. With $X = U$ the adjusted estimate sits on ≈ 1.0: unconfoundedness now holds *exactly*, because the one confounder is fully observed. Adjustment works precisely when its assumption is true.

Part 3. The curve rises smoothly from ≈ 1.0 (no noise) toward the naive value as the proxy degrades — **measurement error in a confounder leaves residual confounding in proportion to the noise**. "We controlled for income" means little if income was self-reported in five broad bins.

Common mistake

Concluding from part 1 that adjustment "does not work". It did exactly what it promises: removed the bias carried by X. The shortfall is the part of U that X never contained.

Extended Exercise A10.2 — Design an observational study [Integrative]

Extended exercise (15 min)

Your company rolled out a **sales-assistant AI** last year: teams could adopt it voluntarily, and adopters' revenue is now 18% higher. The board asks: "so it caused 18%?" Design the honest analysis:

1. Name two plausible confounders and, for each, the *sign* of the bias it induces (use $\gamma\delta$ reasoning).
2. You have monthly revenue per team for two years. Which design from this module fits best, and what exactly would you compute?
3. The rollout was staggered by region for logistics reasons unrelated to performance. What extra design does that enable?
4. List the three robustness checks you would show the board.

Solution — Extended Exercise A10.2

Solution

Part 1. Team quality (good teams adopt tools *and* sell more: $\gamma > 0, \delta > 0 \Rightarrow$ upward bias); desperation adoption (struggling teams grab tools: $\gamma < 0, \delta > 0 \Rightarrow$ downward bias). The 18% could be either side of the truth.

Part 2. **Difference-in-differences** on the monthly panel: adopters vs never-adopters, change from the pre-adoption year to the post-adoption year — with the pre-period trend plot as the parallel-trends evidence.

Part 3. The staggered, logistics-driven rollout is a candidate **instrument**: region-wave assignment shifts adoption timing (relevance, testable) and is plausibly unrelated to team quality (independence — argue it). Compute the Wald ratio on early- vs late-wave regions.

Part 4. (i) Pre-trend plot; (ii) balance table on observables before/ after matching; (iii) sensitivity: how strong an unmeasured confounder ($\gamma\delta$) would have to be to erase the estimate.

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Python lab — run it live

Reproduce every number: `make_figures.py`

Every estimate quoted on these slides is computed by `make_figures.py` in this module's folder, from seeded simulations and the bundled `Wage.csv` — run it and read it alongside the deck:

- the running simulation: naive 2.53 / adjusted 1.35 / matched 1.42;
- the from-scratch logistic propensity model and 1:1 matching (ATT);
- the DiD figure and the IV simulation (OLS 2.33 vs Wald 1.08);
- the `Wage` education gaps (36.3 naive, 35.1 adjusted).

A companion notebook in the course lab format is in preparation; until it lands, the script *is* the lab — it runs top to bottom in the course environment with no extra packages.

The whole module in fifteen lines

```python
import numpy as np
rng = np.random.default_rng(2024)
n = 2000
U = rng.normal(0, 1, n) # unmeasured ability
X = U + rng.normal(0, 0.5, n) # what we actually observe
D = (0.9*U + rng.normal(0, 1, n) > 0).astype(float)
Y = 2 + 1.0*D + 1.5*U + rng.normal(0, 1, n)

naive = Y[D == 1].mean() - Y[D == 0].mean() # 2.53
Z = np.column_stack([np.ones(n), D, X])
adjusted = np.linalg.lstsq(Z, Y, rcond=None)[0][1] # 1.35

print(f"truth 1.00 | naive {naive:.2f} | adjusted {adjusted:.2f}")
# adjustment removes the bias carried by X -- and only that
```

Outline

1 The Problem
2 Omitted-Variable Bias
3 Adjustment & Matching
4 Difference-in-Differences
5 Instrumental Variables
6 Python Lab
7 Summary

Module A10 in one slide

What we accomplished

Four honest ways to chase a causal effect when nobody randomised — each with its assumption stated out loud.

- Naive comparison = effect + selection: on our simulation, 2.53 for a true 1.0.
- Omitted-variable bias = $\gamma\,\delta$: sign and size it without data.
- Adjust / stratify / match on observables: 2.53 $\rightarrow$ 1.35–1.42 — exactly as far as the measured X reaches, never further.
- Difference-in-differences: time-constant confounding cancels; parallel trends is the bet.
- Instrumental variables: the Wald ratio 1.08 recovers the truth without ever seeing U — if the instrument is truly excludable.

Key formulas at a glance

Omitted-variable bias short $= \tau + \gamma\delta$, with $U = \delta_0 + \delta D + v$.

Unconfoundedness $\{Y(1), Y(0)\} \perp D \mid X$ — the licence for all adjustment methods.

Propensity score $e(x) = \Pr(D = 1 \mid X = x)$; balance on $e(X)$ suffices.

SMD balance check $(\bar{x}_1 - \bar{x}_0)/\sqrt{(s_1^2 + s_0^2)/2}$; aim below 0.1.

DiD $(\bar{Y}_{T,\text{aft}} - \bar{Y}_{T,\text{bef}}) - (\bar{Y}_{C,\text{aft}} - \bar{Y}_{C,\text{bef}})$.

Wald / IV $(\bar{Y}_{Z=1} - \bar{Y}_{Z=0})/(\bar{D}_{Z=1} - \bar{D}_{Z=0})$ — ITT scaled by the first stage.

Vocabulary checklist

Term	Meaning
confounder	common cause of D and Y; adjust for it
mediator	lies on the causal path; do not adjust
collider	common *effect*; adjusting creates bias
unconfoundedness	all confounders measured — untestable licence to adjust
propensity score	$\Pr(D = 1 \mid X)$; one scalar summarising all covariates
overlap / common support	both groups present at every propensity value
parallel trends	DiD's assumption: equal untreated time-paths
instrument	shifts D; touches Y only through D; as-good-as-random
first stage	effect of Z on D; weak $\Rightarrow$ unstable IV
complier	unit whose D the instrument actually moves

Decision rules of thumb

- Randomised? Use module A1 and stop. Observational? Start from the diagram, not from the regression.
- Adjust for confounders only — never mediators, never colliders, never post-treatment variables of any kind.
- Before matching or weighting: plot propensity **overlap**. After: show the **balance** table. No overlap $\Rightarrow$ no comparison.
- Panel data and a clean untreated group $\Rightarrow$ prefer DiD; always plot pre-trends.
- An instrument beats adjustment *only if* you can argue exclusion out loud; check the first stage before trusting anything.
- Whatever you estimate, attach the sensitivity question: how big a $\gamma\delta$ would undo it?

Common pitfalls

Don't

1. Read a regression coefficient on observational data as an effect because the p-value is small.
2. Control for everything the dataset offers (mediators and colliders included).
3. Match without checking overlap and balance — or report the estimate without them.
4. Run DiD without a pre-trend plot.
5. Use a clever instrument with a first stage of 0.05.
6. Forget that adjustment on a noisy proxy leaves bias in proportion to the noise.

Self-check questions

Quick self-assessment

1. The naive gap is 2.53 and the truth 1.0. Which two numbers multiply to the bias, and what are they called?
2. Tar deposits in the smoking–disease study: adjust or not, and why?
3. Treated: low 5.0 (n=20), high 9.0 (n=80); control: low 4.0 (n=80), high 8.0 (n=20). Naive and stratified estimates?
4. Control 50 $\to$ 56, treated 48 $\to$ 60: the DiD estimate, and the assumption it rests on?
5. Wald: ITT = 0.43, first stage = 0.40. The IV estimate — and whose effect is it?
6. Matching moved the SMD from 1.06 to 0.00 yet the estimate is still 1.42, not 1.0. What explains the gap?

Answers

(1) γ (confounder $\to$ outcome) $\times$ δ (group gap in the confounder) — omitted-variable bias. (2) Do not adjust: it is a mediator; you would delete the effect's pathway. (3) Naive $8.2 - 4.8 = 3.4$; stratified 1.0. (4) $12 - 6 = 6$; parallel trends. (5) $0.43/0.40 \approx 1.08$; the compliers'. (6) X is a noisy proxy — balance on X is not balance on U; residual confounding remains.

Five things to remember

Module A10 in five lines

1. **No design, no effect.** A coefficient is causal because of an argument about the world, never because of a fit statistic.
2. **Bias $= \gamma\,\delta$.** Two numbers you can reason about even when you cannot measure them.
3. **Adjustment reaches exactly as far as your covariates** — our noisy proxy took 2.53 to 1.4, not to 1.0.
4. **DiD trades confounders for one assumption** — parallel trends — and pre-trends are the only evidence for it.
5. **An instrument imports randomisation**: Wald = ITT / first stage landed on 1.08 without ever seeing the confounder.

What's next

- **Module A1 — Randomised Controlled Trials** is this module's prerequisite and its benchmark: every method here is an attempt to reconstruct, by argument, what randomisation buys by design.
- **Module A2 — Explainable AI with Shapley Values**: its Pitfall 2 ("an explanation is not a cause") is precisely this module's subject — now you have the vocabulary to say why.
- **Module A7 — Multiple Testing**: subgroup-hunting for the segment where the effect "works" is a multiplicity problem before it is a causal one.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Deriving the OVB formula** — three lines of Chapter 3 algebra; moved out because the result, not the derivation, is what the module uses.
- **ATT vs ATE** — which average the estimators actually target; moved out because the distinction matters only once the main ideas sit.
- **Two-stage least squares** — the regression form of the Wald estimator, for continuous instruments and controls.

Deriving the omitted-variable-bias formula

Long model: $Y = \beta_0 + \tau D + \gamma U + \varepsilon$; auxiliary: $U = \delta_0 + \delta D + v$ with $v \perp D$ by construction. Substitute:

$$Y = \beta_0 + \tau D + \gamma(\delta_0 + \delta D + v) + \varepsilon = \underbrace{(\beta_0 + \gamma\delta_0)}_{\text{new intercept}} + \underbrace{(\tau + \gamma\delta)}_{\text{short-regression slope}} D + \underbrace{(\gamma v + \varepsilon)}_{\text{error, } \perp D}.$$

The composite error is uncorrelated with D, so OLS on D alone estimates $\tau + \gamma\delta$ consistently — consistently *wrong*, by exactly the bias term.

Why this is worth three lines

Every "controlling for X changed my coefficient" moment in Chapter 3 is this identity in motion. It also powers sensitivity analysis: solve $\gamma\delta = \hat{\tau}_{\text{naive}} - 0$ to find how strong a confounder must be to explain your whole estimate.

ATT, ATE, and what each estimator targets

Estimand	Question	Natural estimator
ATE $= E[Y(1) - Y(0)]$	effect for a random unit	stratification weighted by stratum size
ATT $= E[Y(1) - Y(0) \mid D = 1]$	effect for those treated	matching treated to controls
LATE	effect for compliers	Wald / IV

Why the distinction matters

Under treatment-effect *heterogeneity* these differ. Our matching estimator reproduced each treated unit's counterfactual — an ATT. The IV answered for compliers only — a LATE. Neither is wrong; each answers a different policy question ("should we keep the programme for its users?" vs "what if we enrolled everyone?").

Two-stage least squares (2SLS)

Algorithm

1. **First stage:** regress D on Z (and controls X); keep fitted $\hat{D}$.
2. **Second stage:** regress Y on $\hat{D}$ (and the same X); the coefficient of $\hat{D}$ is $\hat{\tau}_{\text{2SLS}}$.

How to read this

$\hat{D}$ keeps only the part of treatment variation that the instrument explains — the part with no selection in it. With one binary Z, no controls, 2SLS *equals* the Wald ratio exactly. Report the first-stage F: below ~ 10, the instrument is too weak to trust (our simulation's first stage of 0.40 on $n = 5000$ is comfortably strong).

Common mistake

Running the two stages by hand and using the second stage's printed standard errors — they ignore that $\hat{D}$ was estimated. Real implementations (`linearmodels.IV2SLS`) correct this.